How Costly Are Cartels?

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Any opinions and conclusions expressed in this discussion do not necessarily represent the views of the Banque de France.

This paper: What it does

- The cost of markups is both large and rapidly growing (Edmond et al., 2022; Baqaee and Farhi, 2020)
- But what are (i) the sources and (ii) the economic importance of the distortions generating these markups?

  $\longrightarrow$ This paper studies the impact of one source (collusion) on aggregate productivity and welfare

This paper: What it finds

- Model: Extends Atkeson and Burstein (2008)'s static heterogenous-firm model with oligopolistic competition by introducing *collusive behaviors*
  - Idea: Cartel members deviate from their own-profit maximization and overcharge
- Effect on sectoral productivity:

$$\hat{z}_s = \sum_k \omega_{sk} \left(\frac{\mu_s}{\mu_{sk}} - 1 \right) \hat{P}_{sk} + (\rho - 1) \sum_k \omega_{sk} \frac{\mu_s}{\mu_{sk}} \left(\hat{P}_{sk} - \hat{P}_s \right)$$

  - Direct price effect
  - Reallocation effect :
    - ⋆ (+) if cartel is made of producers with *above* average markups → Demand redirected towards less productive firms → Amplifies misallocation
    - ⋆ (-) if cartel is made of producers with *below* average markups → Demand redirected towards more productive firms → Dampens misallocation
- Effect of cartels on productivity not obvious *a priori*

This paper: What it finds

- Quantification exercise:
  - French Competition Authority cases over the period 1994-2007

Table 6: Aggregate Gains from Breaking Down Cartels

Calibrated model:	Competitive	Collusion		
Breaking down:		All cartels	Larger cartels	Smaller cartels
	(1)	(2)	(3)	(4)
Panel A: Aggregate productivity gains, %				
$A_{\text{cartel}} \to A_{\text{comp}}$		1.11	0.88	0.23
$A \to A_{\text{eff}}$	2.16	3.67	3.67	3.67
Distance to efficient allocation		–30.34	–24.08	–6.15
Panel B: Aggregate welfare gains				
$\mathcal{M}_{\text{cartel}} \to \mathcal{M}_{\text{comp}}$ (in pp)		–1.54	–1.16	–0.39
$C_{\text{cartel}} \to C_{\text{comp}}$ (in %)		2.52	1.97	0.54
$K_{\text{cartel}} \to K_{\text{comp}}$ (in %)		4.11	3.16	0.93
$Y_{\text{cartel}} \to Y_{\text{comp}}$ (in %)		2.84	2.20	0.62
$L_{\text{cartel}} \to L_{\text{comp}}$ (in %)		0.53	0.40	0.13
$\mathcal{W}_{\text{cartel}} \to \mathcal{W}_{\text{comp}}$ (in %)		2.00	1.56	0.41
$\mathcal{W} \to \mathcal{W}_{\text{eff}}$ (in %)	4.95	7.83		

  - Negative effects on aggregate productivity (-1.1%) and welfare (-2%)

- Policy takeaways:
  - Antitrust enforcement and competition laws that aim to break down cartels can yield large gains
  - Targeting some cartels (those with high markups firms) would be a first step

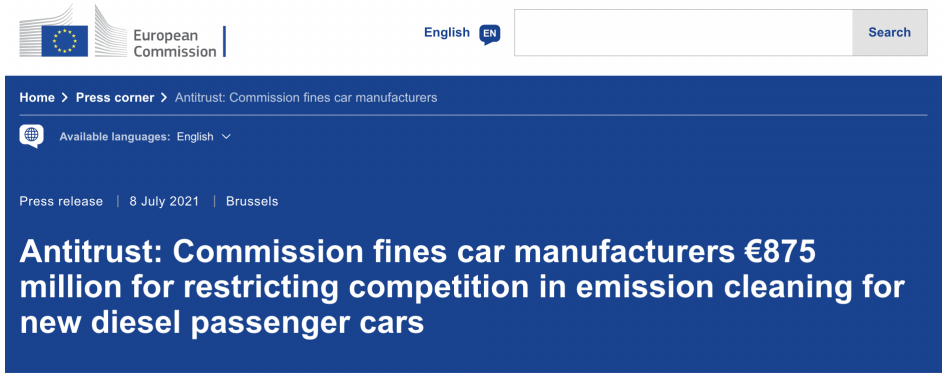
General Comment

- Great paper combining IO and macro to study an important topic
- Analysis is clear, combining theory and empirics
- At a time of industrial policies comeback, provide very useful policy implications for competition laws

Comment #1: France vs. Europe



(a) In the paper



(b) Not in the paper

- *Linoleum floor* cartel
- Price-fixing violation
- Modest (but real) press coverage (Le Monde, 2017)
- 25 years long cartel
- Fine: 302 million €

- *Dieselgate* cartel
- *"Colluding on technical development in the area of nitrogen oxide cleaning"*
- Enormous press coverage of the scandal
- 5 years long cartel
- Fine: 875 million €

Comment #1: France vs. Europe

- Analysis focusing on France but without all French cartels
  - Analysis misses the European Commission DG Comp cases
- Clearly mentionned in the paper: Underestimates the effect of cartels
- Would it be possible to include DG Comp cartels to the analysis?

Comment #2: Cartels and innovation

- Paper mentions the ambiguous dynamic effect of cartels on innovation:
  - Barriers to entry could decrease creative destruction and innovation
  - Cartels, by increasing profits, may boost incumbents' innovation
- Is this positive effect of cartels on incumbents' innovation obvious?
  - Back to *linoleum floor* cartel: "That's why we decided together not to go beyond a certain acoustic performance." (CEO quoted in *Le Monde*)
  - Back to the *Dieselgate* cartel: Collusion to prevent innovation
- What about this "static" and direct effect of cartels on incumbents' innovation, and thus, productivity? Is it taken into account somehow?

Comment #3: Horizontal vs. Vertical

What about vertical arrangements?

- Paper very clear on this point: focuses on horizontal arrangements
- But some cases are typically vertical collusion
  - 2006: Vertical collusion between 16 luxury perfume brands and 3 retailors (46 million € fine)
- To what extent could the model handle these situations?
- GVC considerations?

Comment #4: Markups estimations

- Some recent papers are estimating markups using quantity rather than sales data (Bond et al., 2021; De Ridder et al., 2022)
- Acknowledged in the paper
- Given the time period, not sure such data are available in France (?)